

Supplementary Information for “A universal spectral lower bound on the leave-node-out generalisation error of graph-supervised learning”

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Abstract

This supplementary information accompanies the manuscript *A universal spectral lower bound on the leave-node-out generalisation error of graph-supervised learning* (main text, Sec. 1–8). It provides four appendices: (A) the arbitrary-subspace Pesenson extension lemma used as a black-box reference in the proof of Theorem 2 of the main paper; (B) the transductive Rademacher constants of [El-Yaniv and Pechyony \(2009\)](#) with detailed derivation; (C) full proofs of the negative-transfer (C2) and active-learning (C3) corollaries, with the sampling-restriction correction made explicit; and (D) the empirical reproducibility map across the sibling `cycling-data-lab` repositories. An appendix E compares our finite-population empirical-risk setting to the continuous-Hilbert-space noisy inverse-problem setting of [Bauer and Pereverzev \(2005\)](#), clarifying why their minimax theory does not directly apply to our problem.

A Arbitrary-subspace Pesenson reconstruction lemma

The proof of Theorem 2 (main text, Sec. 5) cites the sampling-quality-regularised reconstruction lemma of [Pesenson \(2008\)](#)/[Anis et al. \(2016\)](#) as a black-box, with the extension from bandlimited subspaces $V_K := \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_K)$ to arbitrary feature subspaces $\mathcal{S} = \text{col}(\mathbf{B})$ attributed to [Chepuri and Leus \(2018\)](#) and [Tanaka and Eldar \(2020\)](#). For self-containment, we record the statement and a 3-line proof.

Lemma A.1 (Subspace Pesenson reconstruction). *Let $\mathbf{B} \in \mathbb{R}^{N \times d}$ be any orthonormal basis matrix ($\mathbf{B}^\top \mathbf{B} = \mathbf{I}_d$) for a closed linear subspace $\mathcal{S} \subseteq \mathbb{R}^N$. Let $T \subset V$ with $\sigma_T(\mathcal{S}) := \lambda_{\min}(\mathbf{B}^\top \mathbf{P}_T \mathbf{B}) > 0$, where $\mathbf{P}_T$ is the diagonal projector onto T . For any $\mathbf{y} \in \mathbb{R}^N$ and any $\tau > 0$, the Tikhonov-regularised reconstruction*

$$\hat{\mathbf{y}}_\tau := \mathbf{B}(\mathbf{B}^\top \mathbf{P}_T \mathbf{B} + \tau \mathbf{I}_d)^{-1} \mathbf{B}^\top \mathbf{P}_T \mathbf{y} \tag{A.1}$$

satisfies

$$\|\hat{\mathbf{y}}_\tau - \mathbf{P}_{\mathcal{S}} \mathbf{y}\|^2 \leq \frac{\tau^2}{(\sigma_T(\mathcal{S}) + \tau)^2} \|\mathbf{P}_{\mathcal{S}} \mathbf{y}\|^2 + \frac{1}{(\sigma_T(\mathcal{S}) + \tau)^2} \|\mathbf{P}_T \mathbf{P}_{\mathcal{S}^\perp} \mathbf{y}\|^2. \tag{A.2}$$

The bound does not depend on $\mathbf{B}$ being a basis of Laplacian eigenvectors; the classical [Pesenson \(2008\)](#) statement is the specialisation $\mathbf{B} = \mathbf{U}_K$.

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Proof. Let $\mathbf{M} := \mathbf{B}^\top \mathbf{P}_T \mathbf{B} \in \mathbb{R}^{d \times d}$, symmetric PSD with $\lambda_{\min}(\mathbf{M}) = \sigma_T(\mathcal{S}) =: \sigma$. Let $\mathbf{c}^* := \mathbf{B}^\top \mathbf{y}$ so that $\mathbf{P}_{\mathcal{S}} \mathbf{y} = \mathbf{B} \mathbf{c}^*$. Decompose $\mathbf{y} = \mathbf{P}_{\mathcal{S}} \mathbf{y} + \mathbf{P}_{\mathcal{S}^\perp} \mathbf{y}$ and compute

$$\mathbf{B}^\top \mathbf{P}_T \mathbf{y} = \mathbf{B}^\top \mathbf{P}_T \mathbf{B} \mathbf{c}^* + \mathbf{B}^\top \mathbf{P}_T \mathbf{P}_{\mathcal{S}^\perp} \mathbf{y} = \mathbf{M} \mathbf{c}^* + \mathbf{B}^\top \mathbf{P}_T \mathbf{P}_{\mathcal{S}^\perp} \mathbf{y}.$$

Hence the Tikhonov estimate in coefficient space is $\hat{\mathbf{c}}_\tau = (\mathbf{M} + \tau \mathbf{I})^{-1} \mathbf{B}^\top \mathbf{P}_T \mathbf{y}$, and

$$\begin{aligned} \hat{\mathbf{c}}_\tau - \mathbf{c}^* &= [(\mathbf{M} + \tau \mathbf{I})^{-1} \mathbf{M} - \mathbf{I}] \mathbf{c}^* + (\mathbf{M} + \tau \mathbf{I})^{-1} \mathbf{B}^\top \mathbf{P}_T \mathbf{P}_{\mathcal{S}^\perp} \mathbf{y} \\ &= -\tau (\mathbf{M} + \tau \mathbf{I})^{-1} \mathbf{c}^* + (\mathbf{M} + \tau \mathbf{I})^{-1} \mathbf{B}^\top \mathbf{P}_T \mathbf{P}_{\mathcal{S}^\perp} \mathbf{y}. \end{aligned}$$

Using $\|(\mathbf{M} + \tau \mathbf{I})^{-1}\|_{\text{op}} \leq 1/(\sigma + \tau)$ and $\|\mathbf{B}^\top\|_{\text{op}} = 1$,

$$\|\hat{\mathbf{c}}_\tau - \mathbf{c}^*\|^2 \leq \frac{\tau^2}{(\sigma + \tau)^2} \|\mathbf{c}^*\|^2 + \frac{1}{(\sigma + \tau)^2} \|\mathbf{P}_T \mathbf{P}_{\mathcal{S}^\perp} \mathbf{y}\|^2.$$

Since $\mathbf{B}$ is an isometry, $\|\hat{\mathbf{y}}_\tau - \mathbf{P}_{\mathcal{S}} \mathbf{y}\| = \|\hat{\mathbf{c}}_\tau - \mathbf{c}^*\|$ and $\|\mathbf{c}^*\| = \|\mathbf{P}_{\mathcal{S}} \mathbf{y}\|$, yielding (A.2). $\square \quad \square$

Remark A.2 (Where Lemma A.1 would enter the main-paper proof). The main-paper proof of Theorem 2 does *not* actually invoke Lemma A.1, because the witness predictor used there is $f_{\mathcal{H}_d}^*$ (the population-risk minimiser), which requires no Tikhonov regularisation. The lemma is preserved here as a reference because: (i) earlier drafts of the program (`notes/01_universal_bound_proof.tex`, since superseded) routed through Pesenson-ridge as the witness; (ii) it is the natural substitute when $\mathbf{P}_{\mathcal{S}} \mathbf{y} \notin \mathcal{H}_d$ (non-realisable case), treated in `notes/03_non_realisable_resolution.tex`; (iii) it is the foundation of the leverage-score active-learning analysis (C3).

B Transductive Rademacher constants

The slack of Theorem 2 of the main paper relies on the transductive Rademacher bound of El-Yaniv and Pechyony (2009, Theorem 1), which we restate here in our notation.

Theorem B.3 (Transductive Rademacher bound). *Let $N, n, u = N - n$ as in (main text, Sec. 3). Let $\ell : \mathbb{R} \rightarrow [0, B]$ be a loss function and $\mathcal{H}_d \subseteq \mathbb{R}^N$ a hypothesis class. Define the transductive Rademacher complexity*

$$\mathcal{R}_n^{\text{trans}}(\mathcal{H}_d) := \frac{1}{n} \left(\frac{1}{n} + \frac{1}{u} \right)^{-1/2} \mathbb{E}_\sigma \left[\sup_{f \in \mathcal{H}_d} \sum_{v \in V} \sigma_v f(v) \right], \quad (\text{B.3})$$

where σ_v are i.i.d. taking $+1$ w.p. $p = n/N$, -1 w.p. p , and 0 w.p. $1 - 2p$. Then for any learner $\hat{f}$ outputting predictors in $\mathcal{H}_d$, with probability $\geq 1 - \eta$ over the uniformly random split (T, T^c) of V at $|T| = n$,

$$L_{T^c}(\hat{f}) \leq L_T^\pi(\hat{f}) + 2\mathcal{R}_n^{\text{trans}}(\ell \circ \mathcal{H}_d) + B \sqrt{\frac{(n+u) \log(1/\eta)}{nu}} \cdot (11/4 + \sqrt{\log(1/\eta)/2}), \quad (\text{B.4})$$

where $L_T^\pi(\hat{f}) := \frac{1}{N} \sum_{v \in T} \ell(\hat{f}(v) - y_v)/\pi(v)$ is the importance-weighted training loss (which under uniform sampling reduces to $L_T(\hat{f}) \cdot N/n$).

The constant $11/4 + \sqrt{\log(1/\eta)/2}$ is bounded by ≈ 5.05 for $\eta = 0.05$, which is the value used in the main paper.

The key structural property is that the bound *directly* controls $L_{T^c} - L_T^\pi$ without an intermediate population-risk term L_V . Routing through L_V would incur a factor of $\rho(\Pi) = N/(N - n)$ on the slack (transduction identity), inflating the slack by $\sqrt{\rho(\Pi) - 1}$ relative to (B.4); see `notes/04_sharp_transductive_constant.tex` §1–3 for the detailed comparison.

Lemma B.4 (Talagrand contraction for the transductive Rademacher). *For the $2M$ -Lipschitz square loss truncated to $[-M, M]$ and any hypothesis class $\mathcal{H}_d \subseteq [-M, M]^N$,*

$$\mathcal{R}_n^{\text{trans}}(\ell \circ \mathcal{H}_d) \leq 4M \cdot \mathcal{R}_n^{\text{trans}}(\mathcal{H}_d).$$

Proof. Talagrand (1996, §4.2) for the standard Rademacher, extended to the transductive case by an analogous symmetrisation argument (El-Yaniv and Pechyony, 2009, §5; the only change is that the σ_v take value 0 on the $1 - 2p$ portion of V). $\square$

C Negative-transfer corollary: technical conditions

We make explicit the sampling-restriction correction of the negative-transfer corollary (C2 in the main paper, Remark on “sampling-restriction correction”).

C.1 Setup

Let $V = V_{\text{src}} \cup V_{\text{tgt}}$ disjoint, $|V_{\text{src}}| = n_s$, $|V_{\text{tgt}}| = n_t$, $N = n_s + n_t$. Let $\mathcal{H}_d$ be a closed linear subspace of $\mathbb{R}^N$ of dimension d , with orthonormal basis $\mathbf{B} \in \mathbb{R}^{N \times d}$.

Define the source-set and target-set sampling-quality scalars

$$\sigma_{V_{\text{src}}}(\mathcal{H}_d) := \lambda_{\min}(\mathbf{B}^\top \mathbf{P}_{V_{\text{src}}} \mathbf{B}), \quad \sigma_{V_{\text{tgt}}}(\mathcal{H}_d) := \lambda_{\min}(\mathbf{B}^\top \mathbf{P}_{V_{\text{tgt}}} \mathbf{B}). \quad (\text{C.5})$$

C.2 Identifiability of the source ERM

Lemma C.5 (Source identifiability). *If $\sigma_{V_{\text{src}}}(\mathcal{H}_d) > 0$ (i.e., V_{src} is a uniqueness set for $\mathcal{H}_d$), then the source-ERM*

$$\hat{\mathbf{c}}_s := \arg \min_{\mathbf{c} \in \mathbb{R}^d} \frac{1}{n_s} \sum_{v \in V_{\text{src}}} (\langle \mathbf{B}(v, \cdot), \mathbf{c} \rangle - y_v)^2$$

is unique and $\hat{\mathbf{c}}_s = (\mathbf{B}^\top \mathbf{P}_{V_{\text{src}}} \mathbf{B})^{-1} \mathbf{B}^\top \mathbf{P}_{V_{\text{src}}} \mathbf{y}$.

Proof. Standard OLS on the design matrix $\mathbf{B}[V_{\text{src}}, :] \in \mathbb{R}^{n_s \times d}$, which has rank d iff $\sigma_{V_{\text{src}}}(\mathcal{H}_d) > 0$. $\square$

C.3 Inner-product approximation on V_{tgt}

Lemma C.6 (Restriction-corrected orthogonality). *Let $f, g \in \mathcal{H}_d$. Then*

$$\left| \langle f, g \rangle_{V_{\text{tgt}}} - \frac{n_t}{N} \langle f, g \rangle \right| \leq M^2 \cdot \left(1 - \sigma_{V_{\text{tgt}}}(\mathcal{H}_d) \frac{N}{n_t}\right)_+, \quad (\text{C.6})$$

where the subscript-+ denotes positive part. In particular, if V_{tgt} is a uniqueness set ($\sigma_{V_{\text{tgt}}} \cdot N/n_t \geq 1$), the correction vanishes and $\langle f, g \rangle_{V_{\text{tgt}}} = (n_t/N) \langle f, g \rangle$.

This lemma is what makes the negative-transfer proof (main paper, Theorem 6) clean: under the assumption that V_{tgt} is a uniqueness set for $\mathcal{H}_d$ (which is implicit in evaluating any non-degenerate target loss), the orthogonality of source and target projections under spectral disjointness translates from the global $\mathbb{R}^N$ inner product to the restricted V_{tgt} inner product.

Remark C.7 (When does V_{tgt} fail to be a uniqueness set?). V_{tgt} fails to be a uniqueness set for $\mathcal{H}_d$ iff $\dim(\mathcal{H}_d) > n_t$, i.e., the target sample is smaller than the hypothesis class dimension. In this regime the target ERM is not identifiable, and the negative-transfer phenomenon is masked by the broader unidentifiability of any prediction on V_{tgt} (regardless of source). We assume $n_t \geq d$ throughout the corollary; in practice this is mild (it requires more target examples than features), and when it fails the more fundamental identifiability question must be addressed first.

D Reproducibility map

The empirical content of the main paper ((main text, Sec. 8), Table 1) is reproducible from publicly archived scripts in the sibling `cycling-data-lab` repositories. We provide a per-row mapping.

Main-paper Table 1 row	Repository	Script (experiments/)
8 MatBench tasks (mp_e_form, mp_gap, etc.)	materials-applicability-bound	d11_predictive_multitask
matbench_expt_gap	materials-applicability-bound	d11b_expt_gap.py
MovieLens user-rating pilot	materials-applicability-bound	d01_movielens_pilot.py
QSAR pilot	materials-applicability-bound	d04_qsar_pilot.py
AFLOW bulk / shear modulus pilots	materials-applicability-bound	d03_aflow_pilot.py
Bike-share LSO panel ($\Delta R^2 \in [0.06, 0.55]$)	bikeshare-demand-forecasting	d19_lso_rigorous.py
Mobility commune panel (in preparation)	mobility-applicability-bound	(in preparation)
Falsifiability F1 (Spearman exact $p = 4.96 \times 10^{-5}$)	materials-applicability-bound	d20_exact_permutation_t
Falsifiability F2 (partial Spearman, bootstrap CI)	materials-applicability-bound	d13_bootstrap_ci.py
Falsifiability F3 (CIG, shuffled- k NN null)	materials-applicability-bound	d25_knn_sensitivity.py,
Spectrum decomposition (Magpie / chem graph)	materials-applicability-bound	d05_rspec_matbench.py, d
CHGNet encoder discrimination	materials-applicability-bound	d24_chgnet_structural_e
Figure 1 (spectral-projection picture)	structural-bounds-framework	d01_spectral_projection.

Table D.1: Per-row reproducibility map for the main-paper empirical section. Every value is regenerable from raw open data via the indicated script, using SEED = 42 and Python 3.12 with the `requirements.txt` of the corresponding repository.

D.1 Random seeds and finite-population corrections

The master random seed is pinned to SEED = 42 in every stochastic script. Per-seed sensitivity is documented in `d18_shuffled_knn_null.py` (10 realisations) and `d25_knn_sensitivity.py` (8 tasks $\times$ 3 k values). The headline Spearman $\rho = +1.000$ is robust to the choice of k across $k \in \{5, 10, 20\}$ (see `outputs/d25_knn_sensitivity_full.json`).

Finite-population corrections in the without-replacement Hoeffding inequality (Serfling 1974) are made explicit in the proofs of Theorems 1 and 2 of the main paper; the corresponding empirical correction factor $1 - (n - 1)/N$ is applied in `d13_bootstrap_ci.py`.

E Comparison to noisy continuous inverse-problem theory

A natural question is why our framework does not invoke the classical minimax theory of noisy linear inverse problems on continuous Hilbert spaces (Bauer and Pereverzev, 2005; Cavalier, 2008). We summarise the mismatch.

Setting comparison.

Feature	Bauer–Pereverzev 2005	Present paper
Domain	continuous Hilbert space	finite graph V , $ V = N$
Forward operator	compact linear operator $A : X \rightarrow Y$	sample restriction $\mathbf{P}_T : \mathbb{R}^N \rightarrow \mathbb{R}^n$
Noise	Gaussian $\sigma\xi$ on observations	none (deterministic target)
Source of randomness	noise ξ	random sub-sample T
Estimand	full signal $f \in X$	held-out loss $L_{T^c}(\hat{f})$
Minimax-optimal estimator	Tikhonov, $\tau \sim \sigma^{2/(2\mu+1)}$	ERM with witness $f_{\mathcal{H}_d}^*$
Rate	$\sigma^{4\mu/(2\mu+1)}$	$M^2/\sqrt{N - n}$ (Berry–Esseen)

Why the rates differ. In Bauer and Pereverzev (2005), the rate $\sigma^{4\mu/(2\mu+1)}$ is the variance-bias trade-off of a Tikhonov regulariser against Gaussian noise of variance σ^2 on observations. In our finite-population setting, the analogous “noise” would be the random restriction $\mathbf{y}|_T$ instead of $\mathbf{y}|_V$, but this is not Gaussian: it is a uniform sub-sample. The Berry–Esseen rate $1/\sqrt{N-n}$ governs the residual fluctuation, which is the right object to bound in this setting; the inverse-problem rate $\sigma^{4\mu/(2\mu+1)}$ does not apply because there is no parametric noise variance to optimise against.

When inverse-problem theory *would* apply. If we augment the framework with noisy observations $y_v^{\text{obs}} = y_v + \xi_v$, $\xi_v \sim \mathcal{N}(0, \sigma_\xi^2)$ i.i.d. on T , the Bauer–Pereverzev rate would govern the additional noise-dependent slack term. This is a natural extension to chemical-precision applications (e.g., DFT energy targets with quantum-Monte-Carlo ground-truth labels carrying explicit noise estimates) but is not operative in the deterministic-target setting of our motivating applications (MatBench, MovieLens, mobility panels).

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